

L J INSTITUTE OF COMPUTER APPLICATIONS
Master of Computer Applications (MCA)
Syllabus for Batch - 2025
Semester: III

Category	Skill Enhancement Course(SEC)			
Course Title	Mobile Computing			
Scheme and Credits	Theory	Tutorial	Lab	Credits
	0	0	6	3
Pre-requisites (if any)	• Basic Knowledge of any Object-Oriented Programming Language (Java / C++ / Python) • Fundamentals of Object-Oriented Programming (OOP) — Classes, Inheritance, Interfaces • Basic Understanding of Data Structures (Lists, Maps, Sets) and Algorithms • Familiarity with REST APIs, JSON format, and Basic Web/App Concepts			

1. Course Objectives:

CO1	Understand the Dart language fundamentals and the Flutter framework architecture
CO2	Build responsive and pixel-perfect UIs using Flutter's widget system and Material 3 design
CO3	Implement state management using Provider, GetX, Riverpod and manage navigation between screens
CO4	Integrate REST APIs, Firebase services (Auth, Firestore, Storage, FCM) and local storage into applications
CO5	Build intelligent and scalable software solutions using MVC architecture and AI-powered services to address real-world application requirements and enhance user experiences. Build and test a complete, market-ready Flutter application as a capstone project

2. Course Contents:

Unit	Course Content	Weightage
Unit I	Dart Fundamentals & Flutter Setup 1. Introduction to Flutter • What is Flutter? • Flutter vs Native vs React Native; • Flutter architecture Widget tree; • Single codebase targeting Android, iOS, web and desktop 2. Environment & Project Setup • Installing Flutter SDK; • Setting up Android Studio & VS Code; • Flutter & Dart plugins; • Android Virtual Device (AVD);	15%

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	• Project structure (pubspec.yaml, folder layout); • Running demo app on emulator and physical device 3. Dart Language Basics • Variables, data types, operators, control flow (if/else, loops, switch); • Functions (named, arrow, optional parameters); Collections — Lists, Sets, Maps 4. Dart OOP Concepts • Classes, constructors, inheritance, abstract classes, interfaces, mixins; • Object-oriented principles applied in Flutter app development 5. Null Safety • Nullable vs non-nullable types; late keyword; null-aware operators (??, ?., !); Importance of null safety in production apps 6. Async Programming • Future, Stream, async/await; Error handling with try/catch/finally; Dart isolates (concept level) 7. Hot Reload & Hot Restart • Difference between Hot Reload and Hot Restart; Practical use during development; Running app on emulator and physical device	
Unit II	Flutter UI & Widget Development 1. Stateless & Stateful Widgets • Widget tree concept; • Everything is a widget; • Stateless vs Stateful Widgets; • Widget lifecycle 2. Core Layout Widgets • MaterialApp, Scaffold, AppBar, Container, Row, Column, Stack, Expanded, Flexible, Padding, Align, Center, SizedBox 3. Content Widgets • Text, RichText, Image (asset/network), Icon, CircleAvatar, Divider 4. List & Grid Widgets • ListView, ListView.builder, GridView, GridView.builder, SingleChildScrollView, RefreshIndicator 5. Card & Form Widgets • Card, ListTile, TextField, TextFormField, Form validation, DropdownButtonFormField, Checkbox, Switch, Slider, DatePicker 6. Material 3 Design & Themes	20%

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	• Introduction to Material 3 (Material You); ColorScheme, ThemeData, Dynamic colors; Dark mode and Light mode support; • Custom fonts using Google Fonts package 7. Responsive UI Design • MediaQuery; • LayoutBuilder; • Adaptive layouts for phone, tablet, and web; • App Icons using flutter_launcher_icons 8. Custom Widgets & Assets • Building reusable custom widgets; • Composition over inheritance; • Adding local images, fonts, and asset files	
Unit III	State Management & Navigation 1. Introduction to state management — why it matters • What is state? Local vs global state; • setState; • InheritedWidget concept; • Introduction to state management — why it matters 2. Provider Package • ChangeNotifier, ChangeNotifierProvider, Consumer widget; • context.watch / context.read; • Multi-provider setup; ◦ GetX basics 3. Riverpod (Introduction) • Why Riverpod over Provider; • StateNotifier; • StateNotifierProvider; • Compile-time safety and developer tooling 4. Navigation & Routing • Navigator push/pop/pushNamed/pushReplacement; • Named routes; • Passing and returning data between screens; • Bottom Navigation Bar; • Drawer Navigation 5. Navigation with State Management (Navigator 2.0) • Route parameters; • Nested navigation; • Deep linking concept; • MVC/MVVM Architecture introduction 6. Animations	20%

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	• Hero animations; • AnimatedContainer; • Page transition animations 7. Performance • Performance optimization	
Unit IV	Backend Integration, Firebase & Storage 1. REST API Fundamentals • HTTP methods (GET, POST, PUT, DELETE); Request/Response structure; • Status codes; • JSON format; • Using Postman for API testing 2. HTTP & Dio Packages • Sending HTTP requests using the http package; • Dio for advanced HTTP (interceptors, cancellation, timeout); • Loading and error handling states in UI 3. JSON Parsing • Manual JSON parsing with dart:convert; • Code generation with build_runner; • Handling nested JSON 4. Firebase Setup & Authentication • Adding Firebase to Flutter; • Firebase Authentication — email/password, Google Sign-In; - Auth state management; • User session persistence and sign out 5. Firebase Storage & Notifications • Storage security rules; • Firebase Cloud Messaging (FCM) — foreground/background notifications; flutter_local_notifications 6. Local Storage • SharedPreferences for simple key-value storage (token management);	25%
Unit V	Security, AI Tools & Capstone Project Part A: App Security & Testing 1. Mobile App Security(Mandatory Docs) • Secure storage (flutter_secure_storage); • API key protection (.env files);	20%

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	• HTTPS enforcement; • Certificate pinning; • GDPR/data privacy compliance Part B: Modern Development Tools & AI-Assisted Development 1. Git & GitHub Introduction to Git and GitHub; Version control basics; Uploading and managing projects on GitHub; Using pub.dev packages 2. AI Tools for Development 3. Using ChatGPT, GitHub Copilot, Gemini, and other free AI tools for UI design, debugging, and code assistance; Best practices for clean code and documentation 4. Gemini / OpenAI API 5. Calling Gemini or OpenAI API from Flutter; Building a simple AI chatbot widget; Flutter AI Toolkit; Camera plugin, Image Picker Part C: Capstone Project 1. Project Proposal Domain selection, feature list, wireframes, tech stack decision 2. UI Prototype All screens built with dummy data; responsive layout 3. Backend Integration Firebase or REST API connected; Authentication implemented 4. Advanced Feature At least one: notifications, maps, camera, or AI chatbot 5. Testing & Optimization Unit tests, widget tests, performance profiling 6. Final Demo & Viva Live app demo + code walkthrough + viva questions	

Unit	Topic	Suggested Free AI Tools / Platforms
Unit I Dart Fundamentals & Flutter Setup [15%]	Introduction to Flutter	ChatGPT (free), Google Gemini

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Unit	Topic	Suggested Free AI Tools / Platforms
	Environment & Project Setup	ChatGPT, GitHub Copilot (free for students), Codeium
	Dart Language Basics	ChatGPT, DartPad (online editor), Codeium
	Dart OOP Concepts	ChatGPT, GitHub Copilot, Codeium
	Null Safety	ChatGPT, Google Gemini, Codeium
	Async Programming	ChatGPT, GitHub Copilot, Phind (free)
	Hot Reload & Hot Restart	ChatGPT, Google Gemini
Unit II Flutter UI & Widget Development [20%]	Stateless & Stateful Widgets	ChatGPT, GitHub Copilot, Codeium
	Core Layout Widgets	GitHub Copilot, Codeium, v0.dev (Vercel)
	Content Widgets	GitHub Copilot, ChatGPT, Codeium
	List & Grid Widgets	GitHub Copilot, ChatGPT, Codeium
	Card & Form Widgets	GitHub Copilot, v0.dev, ChatGPT
	Material 3 Design & Themes	Google Gemini, Material Theme Builder (Google), ChatGPT
	Responsive UI Design	ChatGPT, GitHub Copilot, Google Gemini
	Custom Widgets & Assets	GitHub Copilot, Codeium, ChatGPT
Unit III State Management & Navigation [20%]	State Management Basics	ChatGPT, Google Gemini

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Unit	Topic	Suggested Free AI Tools / Platforms
	Provider Package	GitHub Copilot, ChatGPT, Codeium
	Riverpod (Introduction)	ChatGPT, GitHub Copilot, Phind (free)
	Navigation & Routing	GitHub Copilot, ChatGPT, Codeium
	Navigator 2.0 & MVC/MVVM	ChatGPT, Google Gemini, GitHub Copilot
	Animations	GitHub Copilot, ChatGPT, Rive (free tier)
	Performance Optimisation	ChatGPT, Google Gemini
Unit IV Backend Integration, Firebase & Storage [25%]	REST API Fundamentals	ChatGPT, Postman (free), Google Gemini
	HTTP & Dio Packages	GitHub Copilot, ChatGPT, Codeium
	JSON Parsing	Quicktype.io (free), ChatGPT, GitHub Copilot
	Firebase Setup & Authentication	ChatGPT, Google Gemini, Firebase (free Spark plan)
	Firebase Storage & Notifications	ChatGPT, Google Gemini, Firebase (free Spark plan)
	Local Storage	GitHub Copilot, ChatGPT, Codeium
Unit V Security, AI Tools & Capstone [20%]	Mobile App Security	ChatGPT, Google Gemini
	Git & GitHub	GitHub Copilot, ChatGPT

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Unit	Topic	Suggested Free AI Tools / Platforms
	AI Tools for Development	ChatGPT, Google Gemini, GitHub Copilot, Codeium
	Gemini / OpenAI API Integration	Google AI Studio (free), Gemini API (free tier), ChatGPT
	Capstone Project	GitHub Copilot, ChatGPT, v0.dev, Cursor (free tier)

Program Outcomes (POs)

PO Code	Program Outcomes
PO1	Computational Knowledge
PO2	Problem Analysis
PO3	Design and Development of Solutions
PO4	Modern Tool Usage
PO5	Individual and Team Work
PO6	Professional Ethics
PO7	Communication Skills
PO8	Life-long Learning

CO-PO Mapping Matrix

Mapping Scale:

- 3 = High
- 2 = Moderate
- 1 = Low

COs \ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	1	1	–	–	–	2

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COs \ POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO2.	2	2	3	2	1	–	1	1
CO3	2	3	3	2	1	–	–	1
CO4	2	2	3	3	–	1	–	2
CO5	2	3	3	3	2	1	2	3

Reference Books:

1. Flutter & Dart Cookbook — Richard Rose (O'Reilly, 2023)
2. Beginning App Development with Flutter — Rap Payne
3. Flutter for Beginners (3rd Ed.) — Thomas Bailey & Alessandro Biessek
4. Beginning Flutter — Marco L. Napoli
5. Flutter Complete Reference — Alberto Miola
6. Dart Programming Language Specification — Dart Team